GAF-CNN – Final Result Report

Electrochemical image classification (Gramian Angular Field + Convolutional Neural Network)

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1. Model and input

Item	Value
Model file	CNN_5classes__epo_4000_pat_1000_batch_16_tl.keras
Parameters	20,641
Keras version	3.3.3 (saved 2024-10-10@16:49:11)
Input size	224 x 224 x 3 (GAF image, 100 x 100 RGB on disk)
Number of classes	5
Classes	Benzochinone, Catecolo, Ferro-Ferri, Idrochinone, Idrochinone-Benzochinone
Test pre-processing	rescale 1/255, no augmentation, batch 1, shuffle disabled

2. Dataset used

Kind	Layout	Split	Class	Images
CNN input image	concentrations_based	Test	Benzochinone	3
CNN input image	concentrations_based	Test	Catecolo	3
CNN input image	concentrations_based	Test	Ferro-Ferri	3
CNN input image	concentrations_based	Test	Idrochinone	3
CNN input image	concentrations_based	Test	Idrochinone-Benzochinone	3
CNN input image	concentrations_based	Train	Benzochinone	9
CNN input image	concentrations_based	Train	Catecolo	9
CNN input image	concentrations_based	Train	Ferro-Ferri	9
CNN input image	concentrations_based	Train	Idrochinone	9
CNN input image	concentrations_based	Train	Idrochinone-Benzochinone	9
CNN input image	concentrations_based	Val	Benzochinone	3
CNN input image	concentrations_based	Val	Catecolo	3
CNN input image	concentrations_based	Val	Ferro-Ferri	3
CNN input image	concentrations_based	Val	Idrochinone	3
CNN input image	concentrations_based	Val	Idrochinone-Benzochinone	3
CNN input image	shuffled	Test	Benzochinone	9
CNN input image	shuffled	Test	Catecolo	4
CNN input image	shuffled	Test	Ferro-Ferri	5
CNN input image	shuffled	Test	Idrochinone	4
CNN input image	shuffled	Test	Idrochinone-Benzochinone	3
CNN input image	shuffled	Train	Benzochinone	13
CNN input image	shuffled	Train	Catecolo	14
CNN input image	shuffled	Train	Ferro-Ferri	14
CNN input image	shuffled	Train	Idrochinone	15
CNN input image	shuffled	Train	Idrochinone-Benzochinone	13
CNN input image	shuffled	Val	Benzochinone	3
CNN input image	shuffled	Val	Catecolo	6
CNN input image	shuffled	Val	Ferro-Ferri	5
CNN input image	shuffled	Val	Idrochinone	1
CNN input image	shuffled	Val	Idrochinone-Benzochinone	5
raw GAF output	voltammetry/GAFData	-	Benzochinone	15
raw GAF output	voltammetry/GAFData	-	Catecolo	15
raw GAF output	voltammetry/GAFData	-	Ferro-Ferri	15
raw GAF output	voltammetry/GAFData	-	Idrochinone	15
raw GAF output	voltammetry/GAFData	-	Idrochinone-Benzochinone	15

Total images in data/: 264

3. Overall results

Test split	Images	Correct	Wrong	Accuracy %
shuffled	25	24	1	96.00
concentrations_based	15	15	0	100.00

Combined: 39 of 40 test images classified correctly.

4. Accuracy per class (%)

Class	shuffled	concentrations_based
Benzochinone	88.89	100.00
Catecolo	100.00	100.00
Ferro-Ferri	100.00	100.00
Idrochinone	100.00	100.00
Idrochinone-Benzochinone	100.00	100.00

Average confidence of the predicted class

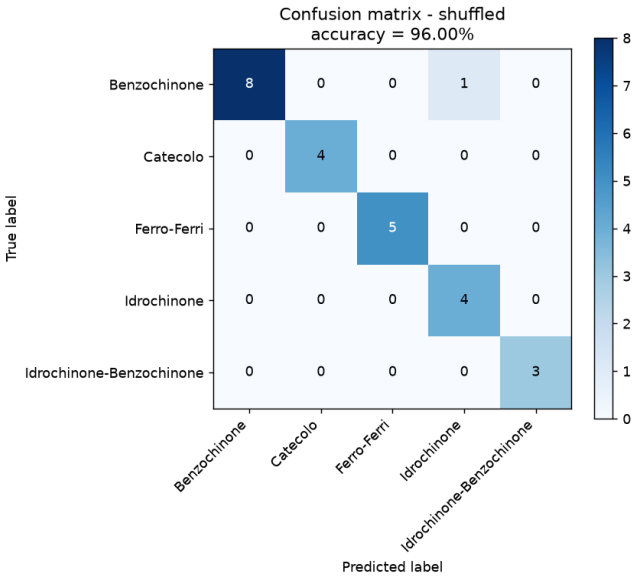
Test split	Correct predictions	Wrong predictions
shuffled	0.9048	0.5987
concentrations_based	0.9511	0.0000

5. Confusion matrices

Rows = true class, columns = predicted class (alphabetical order).

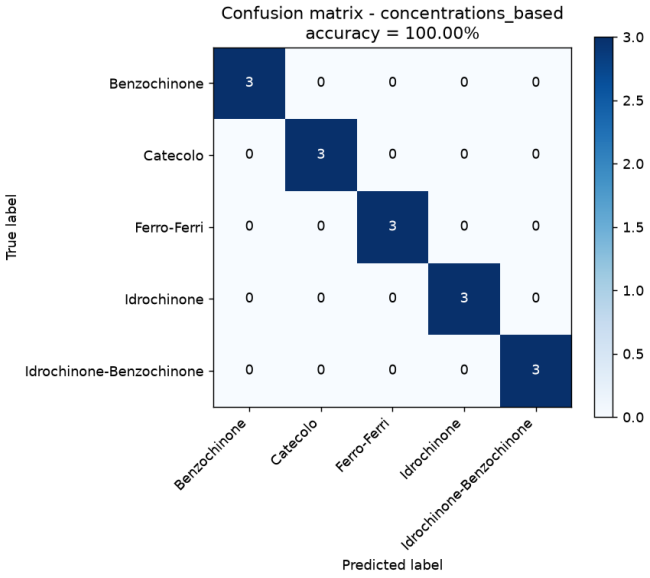
shuffled split

	Benzochinone	Catecolo	Ferro-Ferri	Idrochinone	Idrochinone-B
Benzochinone	8	0	0	1	0
Catecolo	0	4	0	0	0
Ferro-Ferri	0	0	5	0	0
Idrochinone	0	0	0	4	0
Idrochinone-Benzochinone	0	0	0	0	3



concentrations_based split

	Benzochinone	Catecolo	Ferro-Ferri	Idrochinone	Idrochinone-B
Benzochinone	3	0	0	0	0
Catecolo	0	3	0	0	0
Ferro-Ferri	0	0	3	0	0
Idrochinone	0	0	0	3	0
Idrochinone-Benzochinone	0	0	0	0	3



6. Classification reports (precision / recall / f1-score)

shuffled split

```
                         precision    recall  f1-score   support

            Benzochinone     1.0000    0.8889    0.9412         9
                Catecolo     1.0000    1.0000    1.0000         4
             Ferro-Ferri     1.0000    1.0000    1.0000         5
             Idrochinone     0.8000    1.0000    0.8889         4
Idrochinone-Benzochinone     1.0000    1.0000    1.0000         3

                accuracy                         0.9600        25
               macro avg     0.9600    0.9778    0.9660        25
            weighted avg     0.9680    0.9600    0.9610        25
```

concentrations_based split

```
                         precision    recall  f1-score   support

            Benzochinone     1.0000    1.0000    1.0000         3
                Catecolo     1.0000    1.0000    1.0000         3
             Ferro-Ferri     1.0000    1.0000    1.0000         3
             Idrochinone     1.0000    1.0000    1.0000         3
Idrochinone-Benzochinone     1.0000    1.0000    1.0000         3

                accuracy                         1.0000        15
               macro avg     1.0000    1.0000    1.0000        15
            weighted avg     1.0000    1.0000    1.0000        15
```

7. Per image predictions

shuffled split – 25 images

#	File	True class	Predicted class	Conf.	Result
0	0.01mMSPE-Benzochinone_2.png	Benzochinone	Benzochinone	0.6977	OK

#	File	True class	Predicted class	Conf.	Result
1	0.01mMSPE-Benzochinone_3.png	Benzochinone	Benzochinone	0.9017	OK
2	0.1mMSPE-Benzochinone_2.png	Benzochinone	Benzochinone	0.9227	OK
3	0.5mMSPE-Benzochinone_2.png	Benzochinone	Benzochinone	0.7916	OK
4	1mMSPE-Benzochinone_1.png	Benzochinone	Benzochinone	0.7166	OK
5	1mMSPE-Benzochinone_3.png	Benzochinone	Benzochinone	0.9627	OK
6	2mMSPE-Benzochinone_1.png	Benzochinone	Idrochinone	0.5987	WRONG
7	2mMSPE-Benzochinone_2.png	Benzochinone	Benzochinone	0.9547	OK
8	2mMSPE-Benzochinone_3.png	Benzochinone	Benzochinone	0.9576	OK
9	0.01mMSPE-Catecolo_1.png	Catecolo	Catecolo	0.8779	OK
10	0.1mMSPE-Catecolo_1.png	Catecolo	Catecolo	0.9199	OK
11	0.1mMSPE-Catecolo_3.png	Catecolo	Catecolo	0.6279	OK
12	0.5mMSPE-Catecolo_1.png	Catecolo	Catecolo	0.9924	OK
13	0.01mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	Ferro-Ferri	0.5847	OK
14	0.5mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	Ferro-Ferri	0.9929	OK
15	0.5mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	Ferro-Ferri	0.9778	OK
16	2mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	Ferro-Ferri	0.9996	OK
17	2mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	Ferro-Ferri	0.9704	OK
18	0.1mMSPE-Idrochinone_2.png	Idrochinone	Idrochinone	0.9760	OK
19	0.5mMSPE-Idrochinone_2.png	Idrochinone	Idrochinone	0.9986	OK
20	0.5mMSPE-Idrochinone_3.png	Idrochinone	Idrochinone	0.9987	OK
21	2mMSPE-Idrochinone_3.png	Idrochinone	Idrochinone	0.9691	OK
22	H□B1□3SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.9877	OK
23	H□B2□1SPE-Idrochinone-Benzochinone_3.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.9904	OK
24	H□B3□1SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.9468	OK

concentrations_based split – 15 images

#	File	True class	Predicted class	Conf.	Result
0	1mMSPE-Benzochinone_1.png	Benzochinone	Benzochinone	0.7166	OK
1	1mMSPE-Benzochinone_2.png	Benzochinone	Benzochinone	0.9149	OK
2	1mMSPE-Benzochinone_3.png	Benzochinone	Benzochinone	0.9627	OK
3	1mMSPE-Catecolo_1.png	Catecolo	Catecolo	0.9929	OK
4	1mMSPE-Catecolo_2.png	Catecolo	Catecolo	0.9914	OK
5	1mMSPE-Catecolo_3.png	Catecolo	Catecolo	0.9665	OK
6	1mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	Ferro-Ferri	0.9959	OK
7	1mMSPE-Ferro-Ferri_2.png	Ferro-Ferri	Ferro-Ferri	0.9947	OK
8	1mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	Ferro-Ferri	0.9940	OK
9	1mMSPE-Idrochinone_1.png	Idrochinone	Idrochinone	0.9981	OK
10	1mMSPE-Idrochinone_2.png	Idrochinone	Idrochinone	0.9927	OK
11	1mMSPE-Idrochinone_3.png	Idrochinone	Idrochinone	0.9960	OK
12	H□B2□1SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.8397	OK
13	H□B2□1SPE-Idrochinone-Benzochinone_2.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.9206	OK
14	H□B2□1SPE-Idrochinone-Benzochinone_3.png	Idrochinone-Benzochinone	Idrochinone-Benzochinone	0.9904	OK

8. Misclassified images

• shuffled: 2mMSPE-Benzochinone_1.png – true Benzochinone, predicted Idrochinone (confidence 0.5987)
Benzochinone=0.2783 Catecolo=0.0781 Ferro-Ferri=0.0103 Idrochinone=0.5987
Idrochinone-Benzochinone=0.0345

9. Conclusion

The trained CNN classified 39 of 40 test images correctly. Best split: concentrations_based with 100.00% accuracy; shuffled: 96.00%.

The perfect score of the concentrations_based split, which contains every concentration level (0.01, 0.1, 0.5, 1 and 2 mM), shows that the GAF image representation preserves the information needed to recognise the substance independently of its concentration. The single error in the shuffled split is a Benzochinone image at the highest concentration (2 mM) classified as Idrochinone with 0.60 confidence; the two molecules have very similar quinone / hydroquinone redox responses, which explains the confusion.

10. Appendix – full probability matrix

Probability assigned by the CNN to each of the 5 classes, for every test image.

shuffled split

#	File	True	Benzochinon	Catecolo	Ferro-Ferri	Idrochinone	Idrochinone
0	0.01mMSPE-Benzochinone_2.png	Benzochinone	0.6977	0.0514	0.0070	0.2191	0.0247
1	0.01mMSPE-Benzochinone_3.png	Benzochinone	0.9017	0.0249	0.0048	0.0514	0.0173
2	0.1mMSPE-Benzochinone_2.png	Benzochinone	0.9227	0.0284	0.0092	0.0312	0.0084
3	0.5mMSPE-Benzochinone_2.png	Benzochinone	0.7916	0.0577	0.0084	0.1355	0.0068
4	1mMSPE-Benzochinone_1.png	Benzochinone	0.7166	0.0166	0.0310	0.2285	0.0073
5	1mMSPE-Benzochinone_3.png	Benzochinone	0.9627	0.0113	0.0046	0.0118	0.0096
6	2mMSPE-Benzochinone_1.png	Benzochinone	0.2783	0.0781	0.0103	0.5987	0.0345
7	2mMSPE-Benzochinone_2.png	Benzochinone	0.9547	0.0142	0.0044	0.0075	0.0192
8	2mMSPE-Benzochinone_3.png	Benzochinone	0.9576	0.0144	0.0039	0.0072	0.0170
9	0.01mMSPE-Catecolo_1.png	Catecolo	0.0596	0.8779	0.0191	0.0366	0.0069
10	0.1mMSPE-Catecolo_1.png	Catecolo	0.0126	0.9199	0.0407	0.0245	0.0024
11	0.1mMSPE-Catecolo_3.png	Catecolo	0.2350	0.6279	0.0091	0.1191	0.0089
12	0.5mMSPE-Catecolo_1.png	Catecolo	0.0021	0.9924	0.0009	0.0008	0.0037
13	0.01mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	0.1203	0.0292	0.5847	0.2610	0.0047
14	0.5mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	0.0009	0.0008	0.9929	0.0051	0.0001
15	0.5mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	0.0034	0.0015	0.9778	0.0170	0.0003
16	2mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	0.0000	0.0000	0.9996	0.0003	0.0000
17	2mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	0.0081	0.0017	0.9704	0.0187	0.0011
18	0.1mMSPE-Idrochinone_2.png	Idrochinone	0.0102	0.0064	0.0016	0.9760	0.0059
19	0.5mMSPE-Idrochinone_2.png	Idrochinone	0.0002	0.0005	0.0003	0.9986	0.0003
20	0.5mMSPE-Idrochinone_3.png	Idrochinone	0.0002	0.0004	0.0004	0.9987	0.0002
21	2mMSPE-Idrochinone_3.png	Idrochinone	0.0081	0.0106	0.0013	0.9691	0.0109
22	H□B1□3SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	0.0066	0.0037	0.0004	0.0015	0.9877
23	H□B2□1SPE-Idrochinone-Benzochinone_3.png	Idrochinone-Benzochinone	0.0022	0.0037	0.0014	0.0023	0.9904
24	H□B3□1SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	0.0099	0.0203	0.0089	0.0142	0.9468

concentrations_based split

#	File	True	Benzochinon	Catecolo	Ferro-Ferri	Idrochinone	Idrochinone
0	1mMSPE-Benzochinone_1.png	Benzochinone	0.7166	0.0166	0.0310	0.2285	0.0073
1	1mMSPE-Benzochinone_2.png	Benzochinone	0.9149	0.0564	0.0082	0.0101	0.0104
2	1mMSPE-Benzochinone_3.png	Benzochinone	0.9627	0.0113	0.0046	0.0118	0.0096
3	1mMSPE-Catecolo_1.png	Catecolo	0.0014	0.9929	0.0023	0.0010	0.0025
4	1mMSPE-Catecolo_2.png	Catecolo	0.0020	0.9914	0.0023	0.0013	0.0030
5	1mMSPE-Catecolo_3.png	Catecolo	0.0180	0.9665	0.0060	0.0034	0.0061
6	1mMSPE-Ferro-Ferri_1.png	Ferro-Ferri	0.0002	0.0003	0.9959	0.0035	0.0001
7	1mMSPE-Ferro-Ferri_2.png	Ferro-Ferri	0.0003	0.0005	0.9947	0.0044	0.0001

#	File	True	Benzochinon	Catecolo	Ferro-Ferri	Idrochinone	Idrochinone
8	1mMSPE-Ferro-Ferri_3.png	Ferro-Ferri	0.0005	0.0006	0.9940	0.0047	0.0001
9	1mMSPE-Idrochinone_1.png	Idrochinone	0.0003	0.0006	0.0003	0.9981	0.0007
10	1mMSPE-Idrochinone_2.png	Idrochinone	0.0022	0.0020	0.0006	0.9927	0.0025
11	1mMSPE-Idrochinone_3.png	Idrochinone	0.0013	0.0006	0.0014	0.9960	0.0007
12	H□B2□1SPE-Idrochinone-Benzochinone_1.png	Idrochinone-Benzochinone	0.0266	0.0227	0.0837	0.0273	0.8397
13	H□B2□1SPE-Idrochinone-Benzochinone_2.png	Idrochinone-Benzochinone	0.0206	0.0183	0.0144	0.0260	0.9206
14	H□B2□1SPE-Idrochinone-Benzochinone_3.png	Idrochinone-Benzochinone	0.0022	0.0037	0.0014	0.0023	0.9904

11. Generated files

- results/CNN_COMPLETE_OUTPUT.txt
- results/CNN_FULL_REPORT.html
- results/CNN_Final_Report.docx
- results/CNN_Final_Report.txt
- results/README.md
- results/all_generated_images_manifest.csv
- results/all_splits_summary.json
- results/concentrations_based_classification_report.txt
- results/concentrations_based_confusion_matrix.csv
- results/concentrations_based_confusion_matrix.png
- results/concentrations_based_full_output.txt
- results/concentrations_based_predictions.csv
- results/concentrations_based_summary.json
- results/shuffled_classification_report.txt
- results/shuffled_confusion_matrix.csv
- results/shuffled_confusion_matrix.png
- results/shuffled_full_output.txt
- results/shuffled_predictions.csv
- results/shuffled_summary.json